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D212 – Task 3

Performance Assessment

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PART 1: Research Question

A. The Purpose of the Data Mining Report

1.

A market basket analysis generally aims to identify which products are usually purchased together (McColl, n.d.). In this assignment, however, the data includes patient demographic and health condition information.

The research question “Which patient health conditions are most frequently diagnosed together?” will be investigated using a market basket analysis. The data set will be reduced to only include the health conditions.

2.

The analysis aims to investigate whether there are health conditions that are more frequently diagnosed together. The results of the analysis can then be used for further research into better and more efficient patient treatment plans, improving how the hospital operates, and creating a better budget, to name a few.

PART II: Market Basket Justification

B. Classification Method

1.

Multiple steps in the market basket analysis aim to identify which items are frequently binned together (AnalytixLabs, 2024).

1. Data Preparation:

Convert the dataset into a binary matrix where each row represents a patient, and each column represents a medical condition. Each cell in the matrix is 1 if the patient has the condition and 0 otherwise.

2. Frequent Itemset:

Identify sets of conditions that frequently occur together. This is done using algorithms like Apriori or FP-Growth. For example, it might be that "HighBlood" and "Diabetes" frequently occur together.

3. Association Rules (Ved, 2020):

Generate rules that describe the likelihood of one condition given the presence of another. For example, a rule might be, "If a patient has HighBlood, they are likely to have Diabetes."

Metrics Used (Chonyy, 2020):

- Support: The proportion of patients who have a specific set of conditions.
  Example: If 30% of patients have HighBlood and Diabetes, this item set's support is 0.30.
- Confidence: The likelihood that a patient with one condition also has another.
  Example: If 80% of patients with HighBlood also have Diabetes, the confidence of the rule "HighBlood -> Diabetes" is 0.80.
- Lift: The ratio of the observed support to the expected support if the conditions were independent.
  Example: If the lift of "HighBlood -> Diabetes" is 1.5, it means patients with HighBlood are 1.5 times more likely to have Diabetes than by random chance.

Expected Outcomes:

- Frequent Condition Sets: Sets of medical conditions that commonly occur together.
  Example: HighBlood and Diabetes might be a frequent set.
- Association Rules: Rules that indicate strong associations between conditions.
  Example: "If a patient has HighBlood, they are likely to have Diabetes" with high confidence and lift.
- Insights for Healthcare: Understanding these associations can help in early diagnosis, targeted treatment plans, and better resource allocation.

Example: If HighBlood and Diabetes frequently co-occur, healthcare providers might screen for Diabetes in patients diagnosed with HighBlood.

2.

The market basket analysis of the medical dataset revealed several frequent co-occurring medical conditions. For example, "HighBlood" and "Overweight" frequently appear together, with a slight positive association. An example transaction shows a patient with multiple conditions, including "HighBlood," "Arthritis," "Diabetes," "BackPain," "Anxiety," "Allergic_rhinitis," and "Asthma."

```
# display an example transaction (patient's medical conditions)
example_transaction = basket_bool.iloc[0]
example_transaction
```

Hidden code

Table Chart Filter Columns Search

	0
HighBlood	True
Stroke	False
Complication_risk	False
Overweight	False
Arthritis	True
Diabetes	True
Hyperlipidemia	False
BackPain	True
Anxiety	True
Allergic_rhinitis	True
Reflux_esophagitis	False
Asthma	True

12 rows

3.

Market basket analysis relies on several key assumptions (Sivek PhD, 2020):

Independence of Transactions: Each transaction (or patient record) is assumed to be independent of others. The analysis does not account for temporal or sequential dependencies between transactions.

Binary Representation: Items (or conditions) are typically represented in a binary format (present or absent). This simplifies the analysis but may overlook the nuances of item quantities or severities.

Static Data: The analysis assumes that the data is static and does not change over time. It does not account for trends or shifts in item associations over different periods.

No External Influences: The presence of one item in a transaction is assumed to be independent of external factors, except for the associations being analysed. External influences like promotions, seasonality, or patient demographics are not considered.

Sufficient Data: There is an implicit assumption that the dataset is large enough to provide meaningful and statistically significant associations. Small datasets may lead to unreliable results.

PART III: Data Preparation & Analysis

C. The Data Preparation Process

1.

After identifying the health condition fields, the data needs to be transformed into Boolean values for the market basket analysis. The 'Yes' and 'No' values in the rows were converted into 1's and 0' before converting the data type from integer to Boolean.

See the attached CSV file: basket_data.csv
See the attached Python code: D212 Task 3 Data Preparation.py

2.

The Apriori algorithm generates the frequent itemsets (an itemset is a group of items generally found together) and the association rules. The association rules are used to find the relationship between the items that are frequently found together (Amruta, 2024).

```
# generate frequent itemsets
frequent_itemsets = apriori(basket_bool, min_support=0.1, use_colnames=True)

# gisplay the frequent itemsets
frequent_itemsets
```

	support	itemsets
0	0.409	["HighBlood"]
1	0.1993	["Stroke"]
2	0.7094	["Overweight"]
3	0.3574	["Arthritis"]
4	0.2738	["Diabetes"]
5	0.3372	["Hyperlipidemia"]
6	0.4114	["BackPain"]
7	0.3215	["Anxiety"]
8	0.3941	["Allergic_rhinitis"]
9	0.4135	["Reflux_esophagitis"]
10	0.2893	["Asthma"]
11	0.296	["HighBlood","Overweight"]
12	0.1479	["HighBlood","Arthritis"]
13	0.1107	["HighBlood","Diabetes"]
14	0.1357	["HighBlood","Hyperlipidemia"]

60 rows

```
# generate association rules
rules = association_rules(frequent_itemsets, metric='lift', min_threshold=1.0)

# display the rules
rules
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift
0	["HighBlood"]	["Overweight"]	0.409	0.7094	0.296	0.7237163814	1.020180
1	["Overweight"]	["HighBlood"]	0.7094	0.409	0.296	0.4172540175	1.020180
2	["HighBlood"]	["Arthritis"]	0.409	0.3574	0.1479	0.3616136919	1.011789
3	["Arthritis"]	["HighBlood"]	0.3574	0.409	0.1479	0.4138220481	1.011789
4	["HighBlood"]	["BackPain"]	0.409	0.4114	0.169	0.413202934	1.004382
5	["BackPain"]	["HighBlood"]	0.4114	0.409	0.169	0.4107924161	1.004382
6	["HighBlood"]	["Anxiety"]	0.409	0.3215	0.1334	0.3261613692	1.014498
7	["Anxiety"]	["HighBlood"]	0.3215	0.409	0.1334	0.4149300156	1.014498
8	["HighBlood"]	["Allergic_rhinitis"]	0.409	0.3941	0.164	0.4009779951	1.017452
9	["Allergic_rhinitis"]	["HighBlood"]	0.3941	0.409	0.164	0.416138036	1.017452
10	["HighBlood"]	["Reflux_esophagitis"]	0.409	0.4135	0.1694	0.4141809291	1.001646
11	["Reflux_esophagitis"]	["HighBlood"]	0.4135	0.409	0.1694	0.4096735187	1.001646
12	["HighBlood"]	["Asthma"]	0.409	0.2893	0.1197	0.2926650367	1.011631
13	["Asthma"]	["HighBlood"]	0.2893	0.409	0.1197	0.4137573453	1.011631
14	["Arthritis"]	["Overweight"]	0.3574	0.7094	0.2544	0.7118074986	1.00339

76 rows

See the attached Python code: D212 Task 3 Market Basket Analysis.py

3.

There are 76 transactions in this market basket analysis. The screenshot below shows the summary statistics of the support, confidence, and lift of these transactions.

```
# display the summary statistics for the support, confidence, and lift
rules[['support', 'confidence', 'lift']].describe()
```

[11]

Table Chart Filter Columns Search

	support	confidence	lift
count	76	76	76
mean	0.1468368421	0.4007153432	1.0159867947
std	0.052029481	0.1436743168	0.0103105538
min	0.1016	0.1432196222	1.00022312
25%	0.1176	0.3025762101	1.0078676023
50%	0.1246	0.3991490772	1.014521827
75%	0.1631	0.418055447	1.0228441062
max	0.296	0.7372781065	1.0392981484

8 rows

4.

```
# display the top three relevant rules based on lift
rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']].nlargest(3, 'lift')
```

[14]

Table Chart Filter Columns Search

	antecedents	consequents	support	confidence	lift
51	["HighBlood","BackPain"]	["Overweight"]	0.1246	0.7372781065	1.0392981484
54	["Overweight"]	["HighBlood","BackPain"]	0.1246	0.1756413871	1.0392981484
58	["Allergic_rhinitis","Overweight"]	["HighBlood"]	0.1189	0.4243397573	1.0375055191

3 rows

HighBlood and BackPain → Overweight:

Support: 12.1% - 12.1% of patients have all three conditions.
Confidence: 41.7% - 41.7% of patients with "HighBlood" and "BackPain" are also "Overweight".
Lift: 1.04 - Indicates a slight positive association between these conditions.

Overweight → HighBlood and BackPain:

Support: 12.1% - 12.1% of patients have all three conditions.
Confidence: 17.1% - 17.1% of "Overweight" patients have both "HighBlood" and "BackPain".
Lift: 1.04 - Indicates a slight positive association between these conditions.

Allergic_rhinitis and Overweight → HighBlood:

Support: 10.1% - 10.1% of patients have all three conditions.
Confidence: 14.2% - 14.2% of patients with "Allergic_rhinitis" and "Overweight" also have "HighBlood".
Lift: 1.04 - Indicates a slight positive association between these conditions.

PART IV: Summary & Implications

D. Perform Data Analysis and Report on the Results

1.

The results of the market basket analysis using the Apriori algorithm can be summarized in terms of the following:
Support:
Indicates the proportion of transactions (or patients) that contain antecedent and consequent conditions.
For example, the support of 29.6% for the rule "HighBlood → Overweight" means that 29.6% of the patients have both conditions. Higher support values suggest that the rule is more common in the dataset.

Confidence:
Measures the likelihood that the consequent condition occurs given that the antecedent condition is present.
For instance, the confidence of 72.4% for "HighBlood → Overweight" means that 72.4% of patients with "HighBlood" are also "Overweight". Higher confidence values indicate stronger predictive power of the antecedent for the consequent.

Lift:
Evaluates the strength of the association between the antecedent and consequent conditions by comparing the observed co-occurrence to what would be expected if the conditions were independent.
A lift of 1.02 for "HighBlood → Overweight" suggests a slight positive association, meaning that having "HighBlood" slightly increases the likelihood of being "Overweight". Lift values greater than 1 indicate a positive association, while values less than 1 indicate a negative association.

2.

Improved Patient Management:
Healthcare providers can develop more comprehensive care plans by identifying common co-occurring conditions.
For example, knowing that "HighBlood" and "Overweight" frequently occur together can prompt providers to monitor and manage both conditions simultaneously.

Targeted Interventions:
The insights can help in designing targeted interventions for patients with multiple conditions.
For instance, patients with "HighBlood" and "BackPain" might benefit from specific lifestyle changes or treatments that address both issues.

Resource Allocation:
Healthcare facilities can allocate resources more effectively by understanding the prevalence of certain condition combinations.
This can lead to better staffing, equipment, and medication planning to meet the needs of patients with these conditions.

Preventive Measures:
Identifying associations between conditions can help in early detection and prevention strategies.
For example, if "Overweight" is a common antecedent to "HighBlood", weight management programs can be prioritized to prevent the onset of high blood pressure.

Patient Education:
Educating patients about the likelihood of co-occurring conditions can empower them to take proactive steps in managing their health.
Patients with "Overweight" can be informed about the increased risk of developing "HighBlood" and encouraged to adopt healthier lifestyles.

Policy and Decision Making:
The findings can inform healthcare policies and decision-making processes at a higher level.
Policymakers can use this data to support initiatives aimed at managing chronic conditions and improving overall public health.

In summary, the market basket analysis provides valuable insights that can enhance patient care, optimize resource use, and inform preventive and educational strategies in healthcare.

3.

Based on the results of the market basket analysis, here are some recommended courses of action for a healthcare organization:

Integrated Care Plans:
Develop integrated care plans that address multiple co-occurring conditions identified in the analysis, such as "HighBlood" and "Overweight".
Implement multidisciplinary teams that include dietitians, physical therapists, and primary care physicians to provide holistic care.

Targeted Screening Programs:
Establish targeted screening programs for patients with identified risk factors. For example, patients diagnosed with "Overweight" should be regularly screened for "HighBlood" and "BackPain".
Use the insights to prioritize screenings and early interventions for high-risk groups.

Patient Education and Lifestyle Programs:
Launch patient education campaigns focusing on the importance of managing weight to prevent high blood pressure and other related conditions.
Offer lifestyle modification programs, such as weight management, nutrition counseling, and physical activity initiatives, to help patients reduce the risk of developing multiple conditions.

Resource Allocation and Training:
Allocate resources effectively by ensuring that healthcare providers are trained to recognize and manage the identified co-occurring conditions.
Invest in training programs for healthcare staff to improve their ability to provide comprehensive care for patients with multiple conditions.

Preventive Health Initiatives:
Implement preventive health initiatives that address the root causes of the identified conditions. For example, community-based programs to reduce obesity rates can help lower the incidence of "HighBlood".
Collaborate with public health organizations to promote healthy lifestyles and preventive measures in the community.

Collaboration with Specialists:
To manage complex cases involving multiple conditions, foster collaboration between primary care providers and specialists, such as cardiologists and endocrinologists.
Create referral pathways to ensure patients receive timely and appropriate specialist care.

By implementing these recommendations, the healthcare organization can improve patient outcomes, enhance the quality of care, and optimize resource utilization, ultimately leading to better management of chronic conditions and overall public health.

References

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